

Abstraction

IT IS COMMONLY HELD that mathematics began when the perception of three apples was freed from apples and became the integer three. This is an instance of the process of abstraction, but as this word is used in several different but related senses in mathematics, it is important to explain them.

(a) Abstraction as Idealization

A carpenter, using a metal ruler, draws a pencil line across a board to use as a guide in cutting. The line he has drawn is a physical thing; it is a deposit of graphite on the surface of a physical board. It has width and thickness of varying amounts, and in following the edge of the ruler, the tip of the pencil reacts to the inequalities in the surface of the board and produces a line which has warp and corrugation.

Alongside this real, concrete instance of a straight line, there exists the mental idea of the mathematical abstraction of an ideal straight line. In the idealized version, all the accidentals and imperfections of the concrete instance have been miraculously eliminated. There exists an idealized, cosmeticized, verbal description of a straight line: "That which lies evenly with the points on itself" (Euclid: Definition 4) or that curve every part of which is the shortest distance between two of its points. The line is conceived of as potentially extending to infinity on both sides. Perhaps experience in pulling string between the fingers is at the bottom of these descriptions, or experience with the propagation of rays of light, or with folding pieces of paper. But no matter how, idealization has occurred as the end result of a process of perfection. In the end, we may lose or abandon all notions of what a straight line "really is," and merely replace it (if we are at the level of axiomatics) with statements of how the straight line acts or combines. Thus, for example: a straight line is a set of points,

containing at least two points. Two distinct points are contained in one and only one line, etc. (See *The Stretched String*, this chapter.)

Along with the straight line, we arrive at many idealizations and perfections: planes, squares, polygons, circles, cubes, polyhedra, spheres. Some of these, in particular geometrical expositions, will be undefined, e.g. points, lines, planes. Others may be defined in terms of simpler concepts, e.g., the cube. It is well understood that any concrete instance of a cube, say a cubic crystal, will exhibit imperfections, and that any property of the cube inferred mathematically can be verified only approximately by real-world approximations. It is proved in plane geometry that in any triangle the three angle bisectors intersect in a single point; but real-world experience teaches us that no matter how carefully a draftsman draws this figure the bisectors will be only approximately concurrent; the figure will embody a degree of blur or fuzz which the eye will perceive but which the mind will obligingly overlook.

The idealizations just mentioned have proceeded from the world of spatial experience to the mathematical world. Aristotle has described this process by saying (*Metaphysics*, 1060a, 28–1061b, 31) that the mathematician strips away everything that is sensible, for example, weight, hardness, heat, and leaves only quantity and spatial continuity. The making of contemporary mathematical models exhibits updated versions of Aristotle's process. As an example, we quote from a popular book on differential equations (S. L. Ross, Blaisdell, N.Y., 1964, pp. 525–6):

We now make certain assumptions concerning the string, its vibrations, and its surroundings. To begin with, we assume that the string is perfectly flexible, is of constant linear density ρ , and is of constant tension T at all times. Concerning the vibrations, we assume that the motion is confined to the xy plane and that each point on the string moves on a straight line perpendicular to the x axis as the string vibrates. Further, we assume that the displacement y at each point of the string is small compared to the length L and that the angle between the string and the x axis at each

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point is also sufficiently small. Finally, we assume that no external forces (such as damping forces, for example) act upon the string.

Although these assumptions are not actually valid in any physical problem, nevertheless they are approximately satisfied in many cases. They are made in order to make the resulting mathematical problem more tractable. With these assumptions, then, the problem is to find the displacement y as a function of x and t .

The mathematical idealization.

Under the assumptions stated it can be shown that the displacement y satisfies the *partial differential equation*,

$$\alpha^2 \frac{\partial^2 y}{\partial x^2} = \frac{\partial^2 y}{\partial t^2}, \quad (14.20)$$

where $\alpha^2 = T/\rho$. This is the one-dimensional wave equation.

Since the ends of the string are fixed at $x = 0$ and $x = L$ for all time t , the displacement y must satisfy the *boundary conditions*

$$\begin{aligned} y(0, t) &= 0, & 0 \leq t < \infty; \\ y(L, t) &= 0, & 0 \leq t < \infty. \end{aligned} \quad (14.21)$$

At $t = 0$ the string is released from the initial position defined by $f(x)$, $0 \leq x \leq L$ with initial velocity given by $g(x)$, $0 \leq x \leq L$. Thus the displacement y must also satisfy the *initial conditions*.

$$\begin{aligned} y(x, 0) &= f(x), & 0 \leq x \leq L; \\ \frac{\partial y(x, 0)}{\partial t} &= g(x), & 0 \leq x \leq L. \end{aligned} \quad (14.22)$$

This, then, is our problem. We must find a function y of x and t which satisfies the partial differential equation (14.20), the boundary conditions (14.21), and the initial conditions (14.22).

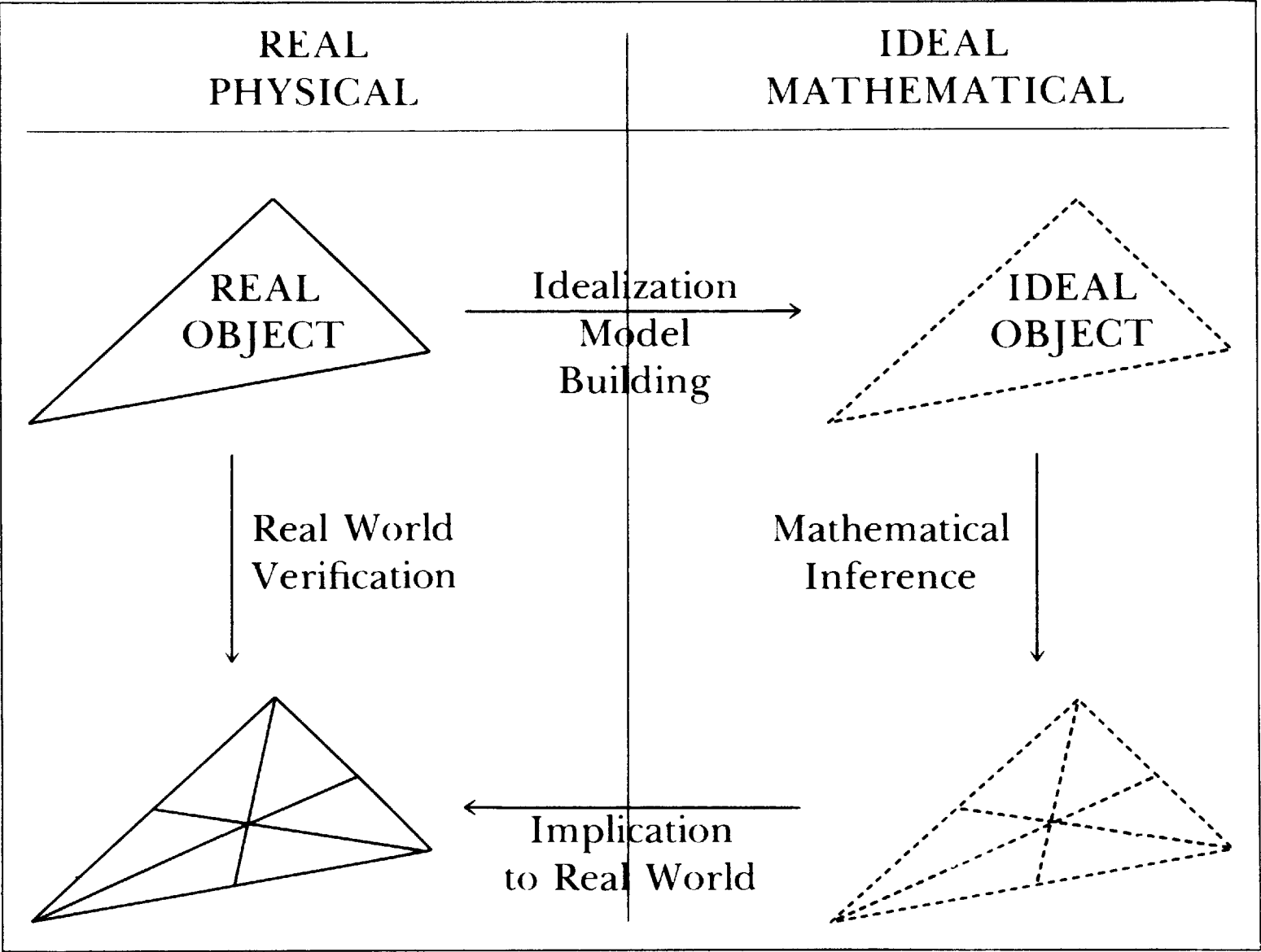
The system of mathematical equations is an idealization of an exceedingly complex set of physical conditions.

The relationship between the real and the ideal is illustrated by the accompanying diagram (though, strictly speaking, we cannot draw the ideal objects on the right side of the diagram).

Intimately related to mathematical idealization is Plato's

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The mathematical idealization



notion of the world of idealized objects. The so-called real world of experience, says Plato, is not real at all. We are like dwellers in a cave, who perceive the shadows of the external world and mistake the shadow for the true thing (*Republic* VII.514–517). The objects of mathematics are all abstract and the Platonic world is the dwelling place of the true circle, and the true square. It is the dwelling place of the true forms, the true perfections, and for this world the language of mathematics is said to provide true descriptions. “Without doubt,” wrote Kepler in 1611, “the authentic type of these figures exists in the mind of God the Creator and shares His eternity.”

(b) Abstraction as Extraction

Four birds are eating bread crumbs in my back yard. There are four oranges on my kitchen table. The very use of the word “four” implies the existence of a process of abstraction wherein a common feature of the birds and of the oranges has been separated out. For each bird there is an orange. For each orange there is a bird, and in this way, there is a one-to-one correspondence between birds and

oranges. Here, in one setting, are objects. There, in another, are abstract numbers, existing, apparently, in isolation from birds or oranges.

“Arithmetic,” says Plato (*Republic* VII.525), “has a very great and elevating effect, compelling the soul to reason about abstract number, and rebelling against the introduction of visible or tangible objects into the argument.”

Today mathematics largely leaves aside the interesting psychohistorical problem of how abstractions come about and concentrates on a set-theoretic description of abstraction formation. The abstract notion of four is, according to Russell and Whitehead (*Principia Mathematica*, vol. I), the set of all sets that can be put into a one-to-one correspondence with the four birds on my lawn.

It will be illuminating to illustrate the process of mathematical abstraction by another example which is at once simple and modern. It is drawn from the theory of abstract graphs. Take a look at the two figures here and ask, what do these figures have in common? At the first glance it may seem that they have nothing in common. The first figure seems to be a series of boxes within boxes, while the second might represent a simplified version of a pearl necklace. There is no doubt, however, that the second is a much simpler figure than the first. Yet there is one very important respect in which these figures are completely identical. Think of the first figure as the plan of a maze or a labyrinth. Starting from the outside, we try to find our way into the innermost chamber. We walk down the corridors more or less at random trying to find a door which will take us one more layer further in, and hoping that we do not

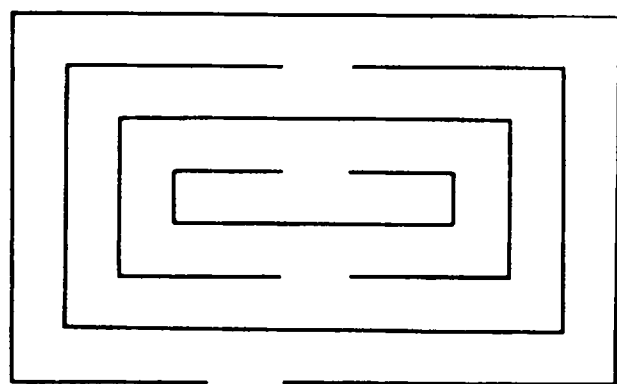


Figure 1.

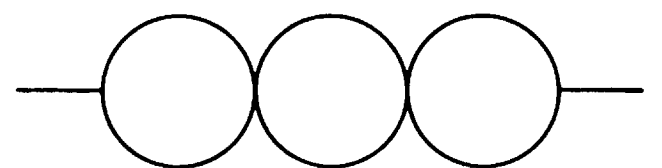


Figure 2.

come back to a spot where we've already been. Once we have made a complete investigation of the labyrinth, we can describe it completely. We may even do it verbally. Suppose that the maze is labeled as in the third figure. Then a description might go as follows:

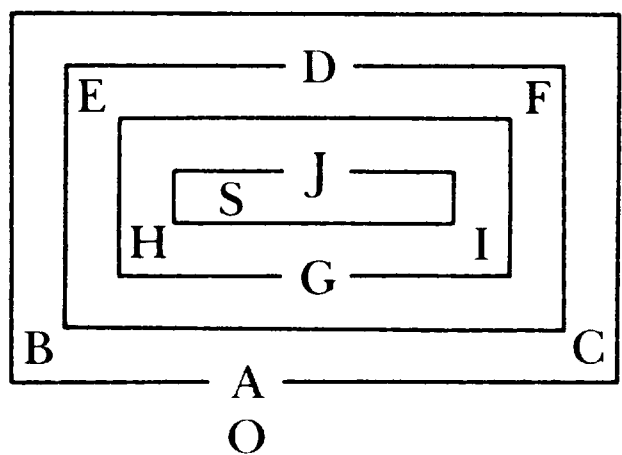
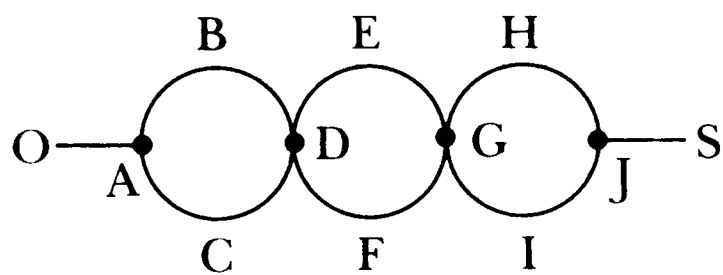


Figure 3.

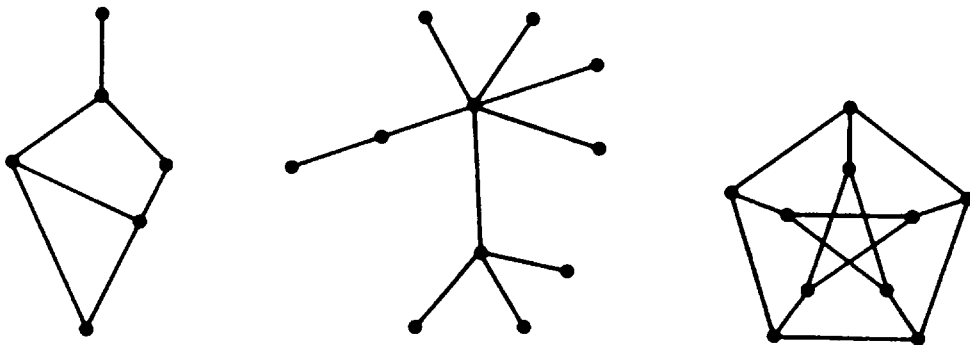
From the outside O go to the door at A. The door at A leads to two halls, B and C, both of which lead to a door D. The door D leads to two halls, E and F, both of which lead to a door G. The door G leads to two halls, H and I, both of which lead to the door J. The door J leads to the innermost chamber S.

Now suppose that we mark Figure 2 as follows.

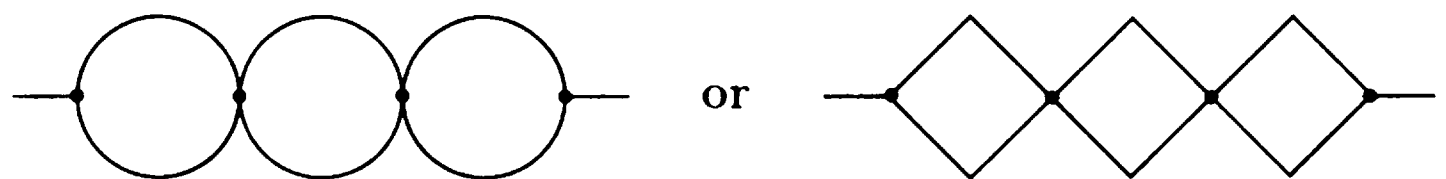


Then you can easily see that the verbal description which applies to the first maze applies also to the above figure as we traverse it from left to right. These two figures are therefore identical in this respect, and the second is a lot simpler to work with conceptually. Naturally, the second figure does not contain as much information about the labyrinth as the first, which may have been an exact floor plan. But if we are interested in the problem of *traversing* the maze, then the second diagram is adequate.

An object such as Figure 2 wherein we are interested only in all possible traversings is known as a *graph*. Additional examples of graphs are drawn here.

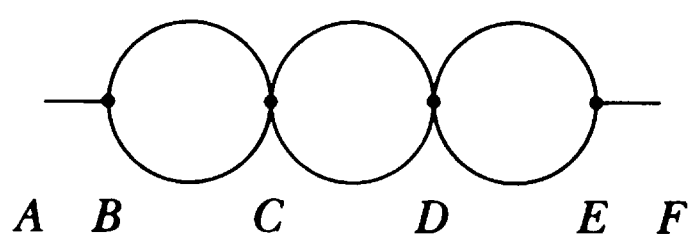


Notice that certain aspects of the geometrical configuration of Figure 2 are inessential. Thus, it makes no difference whether we draw



As regards the traversing process, we can adopt the one which is simplest visually.

The process of abstraction can be continued further, and can depart from geometry completely. In the above graphs, I have marked by small dots the places where alternatives occur. Suppose that I label these decision points with letters, thus:



The lines which connect the decision points are now merely symbolic.

It would serve our purpose as well to write “AB, 1; BC, 2; CD, 2; DE, 2; EF, 1” by way of indicating that one hall connects A with B, two separate halls connect B with C, etc., and there are no other connections. One can even put all this information in a tableau as follows.

	A	B	C	D	E	F
A	0	1	0	0	0	0
B	1	0	2	0	0	0
C	0	2	0	2	0	0
D	0	0	2	0	2	0
E	0	0	0	2	0	1
F	0	0	0	0	1	0

This is a so-called “incidence matrix” of Poincaré. The labyrinth of Figure 1 has been completely described in an arithmetic fashion.

If we are interested in setting up a theory of graphs which describes only the properties of traversings and nothing else, we need nothing more than the information in this matrix.

With this as our raw material, we proceed to make deductions. Although the original inspiration may have been geometric, we have stripped away all the geometric accoutrements, and the theory of abstract graphs is purely combinatorial. The abstract graph emerges as a set of things (nodes) together with a set of relationships between these nodes (paths) which satisfy certain axiomatic requirements. More than this is not necessary.

Further Readings. See Bibliography

J. Weinberg